Homework — 2.1.2 Thinking Ahead — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [3] Completed table:

Item	Input / Output / Precondition
The member's ID scanned at the desk	**Input** (data flowing into the routine)
The updated loan record written to the database	**Output** (a result flowing out)
"The member has no unpaid fines"	**Precondition** (must be true before issuing goes ahead)
The book's barcode scanned at the desk	**Input**
The printed receipt showing the due date	**Output**
"The book is currently on loan to this member"	**Precondition** (must be true before the return routine runs — otherwise a book could be marked returned that was never borrowed, corrupting the loan records)

Mark: 2 correct rows per mark (6 rows → 3 marks). The bracketed reasons are not required.

2. [2] Correct ordering:

Step	Order
The computed timetable is stored in the cache.	**4**
A user requests the London → Manchester timetable.	**1**
The system checks the cache — the timetable is not there (a cache miss).	**2**
A second user requests the same timetable and is served the stored copy straight from the cache.	**6**
The timetable is computed by querying the live database.	**3**
The timetable is returned to the first user.	**5**

Full sequence: request → cache checked (miss) → computed from the database → stored in the cache → returned to the user → next request served from the cache. Mark: all six positions correct = 2; four or five correct = 1.

3. (a) [1] Any one: the expensive computation/database query happens only **once** per day, so pages load faster for every user and server load is greatly reduced.

(b) [2] 1 mark: the cached copy becomes **stale** — if a train is delayed or cancelled during the day, the cache still holds the midnight version. 1 mark: every user is then shown **out-of-date/wrong** times for the

rest of the day because the design never invalidates or refreshes the cached data.

(c) [1] Any one: refresh/invalidate the cache at shorter intervals (e.g. every few minutes); invalidate the cached timetable whenever a delay/cancellation is recorded so it is recomputed; cache only the fixed timetable but fetch live delay information separately.

4. [2] Any two (1 each): faster development — write once, drop into all three games; fewer bugs because the component is already **tested**; consistency — all games save/load the same way; easier maintenance — fix or improve it once and all three benefit.

5. [3] 1 mark each, applied to the order routine: **input** — e.g. items chosen, quantities, delivery address, payment details; **precondition** — e.g. the basket must be non-empty / the items must be in stock / the user logged in *before* payment is taken; **cache** — e.g. the menu/prices, or the user's saved delivery address, stored for fast reuse. Must be applied to the scenario; generic definitions score poorly.

Part B (6 marks)

6. [2] 1 mark keep: e.g. student ID, name, year/class, the subjects/sets they take. 1 mark discard: e.g. eye colour, favourite food, home address detail — irrelevant to scheduling lessons. *(Discard need not be justified here; one keep + one discard.)*

7. [2] 1 mark: an interrupt is a signal sent to the processor that an event needs attention, causing it to pause the current task. 1 mark, any one reason: to respond promptly to I/O (key press, printer ready), errors, or hardware/timer events without constantly polling; the OS saves state, services the interrupt (via the ISR), then resumes.

8. [2] Any two (1 each): clock speed; number of cores; cache size/amount of cache; word length/bus width; pipelining. *(Accept any two valid factors.)*

Total: 14 + 6 = 20 marks